Analysis and Prediction of Factors influencing Nuclear Power Plant Cost

Data 603 – Final Project Presentation

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Introduction

- Our topic is about analysing **numerous factors** which contribute to the **expensiveness** of those **nuclear reactors**. For us, this topic is important because Nuclear is an Eco-friendly source of power generation and should be widely adopted over Coal Power.`
- We are exploring this using **Statistical Analysis.**
- We are interested in **examining** the best fit model which can **predict the cost** based on predictor variables and various interaction terms.

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Our Dataset

- Our dataset is not readily available thus we had to perform web-scrapping and for each country in total 27 countries and each reactor in country we had to collect data.
- Through web-scrapping we collected every variable except:

Cost (Collected manually as it should be accurate)

Efficiency % (Equation : Gross Capacity/Thermal Capacity * 100) – logic added to python script

Duration (End Year – Start Year) – logic added to python script.

C: > Users > Devanshi > WebScrapping > ND.py > ...

```python
elemts_India = ['289','290','291','292','308','286','986','852','853','1059','304','305','306','307','302','303','287','288','295','297','285','296','300','301']
elemts_Japan = ['345','377','389','334','340','341','342','350','351','361','364','336','352','372','373','335','349','362','376','843','842','348','358','374','333','359','365','380','381','367','383','384','366','386','338','357','339','343','
elemts_Ukraine = ['568','569','575','576','573','574','570','571','572','589','577','578','579','584','586','594','595','566','567']
elemts_UK = ['264','265','266','267','231','232','233','234','252','253','254','255','242','243','275','276','248','249','250','251','256','257','259','260','271','272','244','245','269','270','246','247','237','238','235','236','263','261','262
elemts_Spain = ['153','154','155','156','140','139','149','141','150','146']
elemts_Swedan = ['528','538','540','541','530','532','534','535','531','536','537','539','529']
elemts_Koreo = ['411','412','396','397','402','403','413','395','398','399','404','406','394','405','409','410','885','886','887','864','865','883','884','407','408','400','401']
elemts_Finland = ['157','158','159','160','161']
elemts_UAE = ['1050','1051','1052']
elemts_Germany = ['114','85','91','107','86','108','97','123','124','125','126','127','101','104','100','102','89','106','135','94','133','96','93','88','119','121','87','98','122','83','92','90','136']
elemts_Czech = ['78','79','80','81','74','75']
elemts_Pakistan = ['428','429','1044','1045','427','1067','1068']

elemts_Swizerland= ['55','57','58','59','966','56']
elemts_Hungary = ['278','279','280','281']
elemts_Slovakia = ['543','548','549','546','547','550','551']
elemts_SouthAfrica = ['836','837']
elemts_Mexico = ['420','421']
elemts_Romania = ['442','443']
elemts_Iran = ['310']
elemts_Slovenia = ['542']
elemts_Netherlands =['423','422']
elemts_Lithunia = ['415','416']
elemts_Italy = ['327','326','325','318']
elemts_Kazakhstan = ['414']

Country_lst= [elemts_USA,elemts_Russia,elemts_France,elemts_India,elemts_Japan,elemts_Ukraine,elemts_UK,elemts_Spain,elemts_Swedan,elemts_Koreo,elemts_Finland,elemts_UAE,elemts_Germany,elemts_Czech,elemts_Pakistan,elemts_Swizerland,elemts_Hungar
#Country_lst = [elemts_UAE]
df = pd.DataFrame(columns = ['Name','Country','Reactor_Type','Gross_Capacity_MWe','Thermal_Capacity_MWt','Efficiency_Percentage','Starting_Year','End_Year','Duration','Inflation'])
for every_ele in Country_lst:
    for ele in every_ele:
        driver.get(URl+ele)
        print(ele)


        #for number in range(1,2):
        Name = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/h1/span[1]')
        Country = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[2]/div/ul/li[1]/a')
        Reactor_Type = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[2]/td[1]/h5/span')
        Gross_Capacity_MWe = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[4]/td[3]/h5/span')
        Thermal_Capacity_MWt = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[4]/td[4]/h5/span')
        Efficiency_Percentage = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[4]/td[4]/h5/span')
        Starting_Year = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[6]/td[1]/h5/span')
        End_Year = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[6]/td[2]/h5/span')
        Duration = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[6]/td[1]/h5/span')

        Upper = int(Gross_Capacity_MWe[0].text)
        Lower = int(Thermal_Capacity_MWt[0].text)

        str1 = str(Starting_Year[0].text)
        str2 = str(End_Year[0].text)

        Sleep1 = int(str1[8:12])
```


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How does our data look?(Nuclear Data.csv)

- **Response Variable:** Cost
- **Predictor Variables (Total: 10):** Country, Reactor Type, Gross_Capacity_Mwe, Thermal_Capacity_MWt, Efficiency_Percentage, Starting_Year, Duration, Average_inflation_Percentage, Average_Concrete_Price, Average_Steel_Price (Please note : Name of the reactor is unique and having that many categorical variables is not feasible)
- **Total rows:** 261, **Columns:** 12

Cost_Billion_Dollars	Name	Country	Reactor_Type	Gross_Capacity_MWe	Thermal_Capacity_MWt
1.261	ANO-1	UNITED STATES OF AMERICA	PWR	903	2568
1.261	ANO-2	UNITED STATES OF AMERICA	PWR	1065	3026
4.26	BEAVER VALLEY-1	UNITED STATES OF AMERICA	PWR	959	2900

Efficiency_Percentage	Starting_Year	Duration	Average_Inflation_Percentage	Average_Concrete_Price	Average_Steel_Price
35.16	1968	6	0.0488	37.123333	34.33333333
35.19	1968	10	0.0616	41.7	44.7
33.07	1970	6	0.066	44.66237	42.66666667

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Statistical Analysis (Multicolinearity)

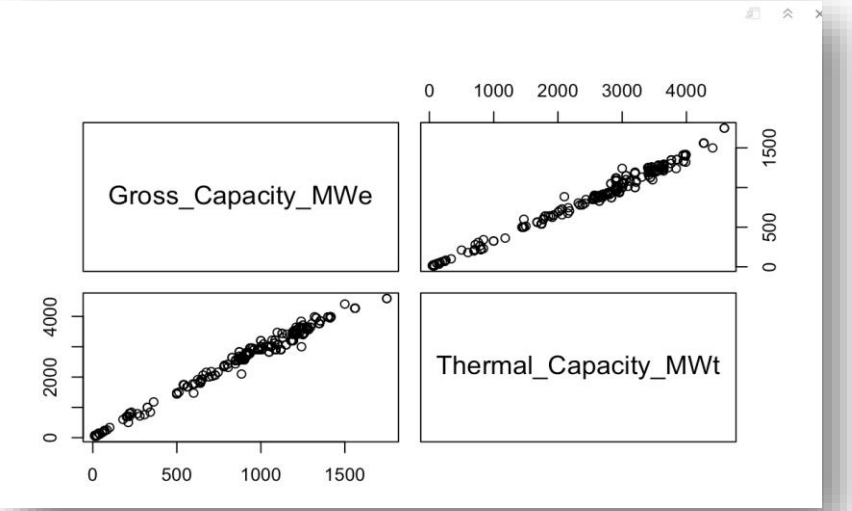
We calculate VIF for our model with all the predictor variables.

```
VIF Multicollinearity Diagnostics

                                  VIF detection
Gross_Capacity_MWe            126.0471         1
Thermal_Capacity_MWt          114.5285         1
Efficiency_Percentage           2.1803         0
Starting_Year                  48.0230         1
Duration                        1.7462         0
Average_Inflation_Percentage    1.0866         0
Average_Concrete_Price         78.4207         1
Average_Steel_Price            52.4073         1

Multicollinearity may be due to Gross_Capacity_MWe Thermal_Capacity_MWt Starting_Year
Average_Concrete_Price Average_Steel_Price regressors

1 --> COLLINEARITY is detected by the test
0 --> COLLINEARITY is not detected by the test
```



Since, VIF value of Gross_Capacity_Mwe > Thermal_Capacity_MWt.
Therefore we removed Thermal_Capacity_MWt from our model.

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Statistical Analysis (Multicolinearity)

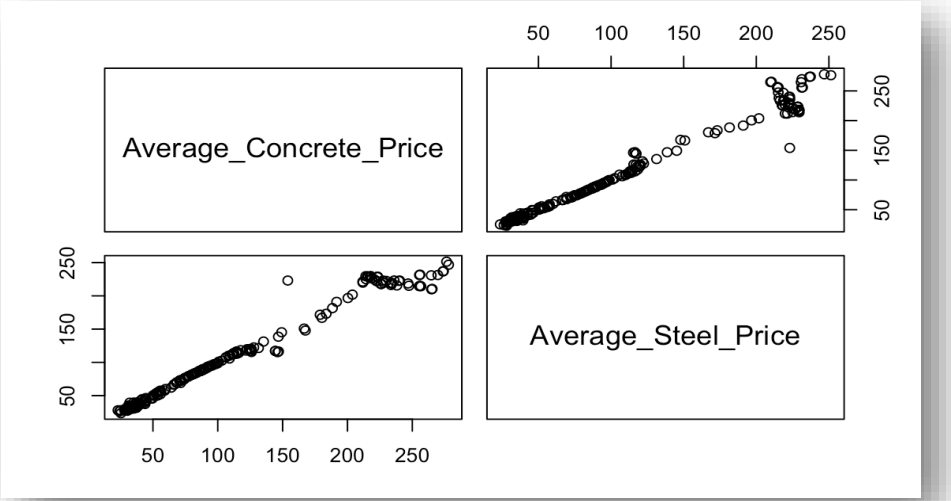
We calculate VIF for our model with all the predictor variables.

```
 VIF Multicollinearity Diagnostics

                                  VIF detection
Gross_Capacity_MWe             1.6770         0
Efficiency_Percentage          1.6309         0
Starting_Year                 46.7577         1
Duration                       1.6575         0
Average_Inflation_Percentage   1.0254         0
Average_Concrete_Price        77.8593         1
Average_Steel_Price           51.4576         1

Multicollinearity may be due to Starting_Year Average_Concrete_Price
Average_Steel_Price regressors

1 --> COLLINEARITY is detected by the test
0 --> COLLINEARITY is not detected by the test
```



Since, VIF value of Average_Concrete_Price > Average_Steel_Price.
Therefore, we removed Average_Steel_Price from our model.

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Statistical Analysis (Multicolinearity)

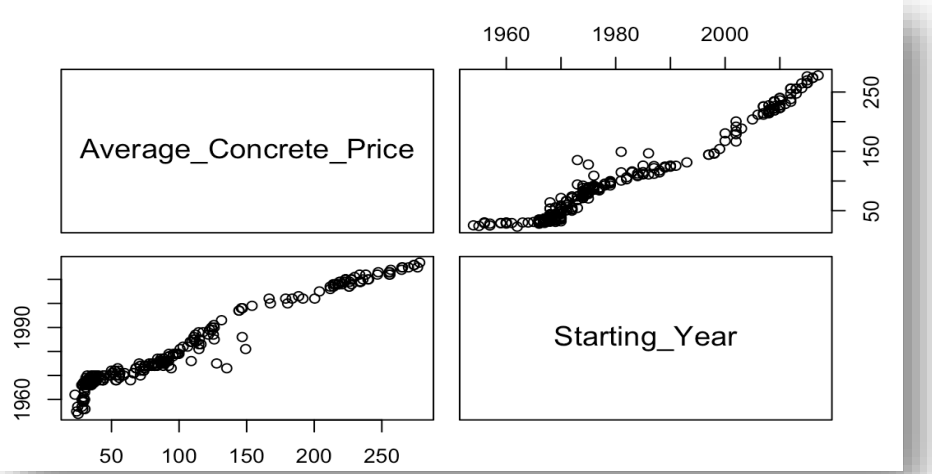
We calculate VIF for our model with all the predictor variables.

```
 VIF Multicollinearity Diagnostics

                                  VIF detection
Gross_Capacity_MWe             1.6767         0
Efficiency_Percentage          1.6247         0
Starting_Year                 44.9093         1
Duration                       1.6144         0
Average_Inflation_Percentage   1.0214         0
Average_Concrete_Price        45.1784         1

Multicollinearity may be due to Starting_Year Average_Concrete_Price regressors

1 --> COLLINEARITY is detected by the test
0 --> COLLINEARITY is not detected by the test
```



Since, VIF value of Average_Concrete_Price > Starting Year.
Therefore we removed Starting Year from our model.

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Model after removal of multicolinearity:

- Country
- Efficiency %
- Duration
- Reactor_Type
- Gross_Capacity_Mwe
- Average_inflation_percentage
- Average_Concrete_Price
- **Total: 7 predictor variables from 10 previously.**

$$new.nuclear.regression.model$$
$$< -lm(Cost_Billion_Dollars \sim factor(Country)$$
$$+ factor(Reactor_Type) + Gross_Capacity_MWe$$
$$+ Efficiency_Percentage + Duration$$
$$+ Average_Inflation_Percentage + Average_Concrete_Price, data$$
$$= nuclear.df)$$

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Statistical Analysis (Global F-test)

- Global F test which tells us if the multiple regression model is useful. To address the overall question, we will test,

$$H0: \beta 1 = \beta 2 = \ldots = \beta 7 = 0$$
$$Ha: \text{ At least one } \beta i \text{ is not zero } (i = 1,2, \ldots, 7)$$

- Our null hypothesis basically means that all of our supposedly Independent variables are not useful at all. While, our Alternative Hypothesis means that at least one of the independent variable is useful in our model.
- Using ANOVA table,

```
Analysis of Variance Table

Model 1: Cost_Billion_Dollars ~ 1
Model 2: Cost_Billion_Dollars ~ factor(Country) + factor(Reactor_Type) +
    Gross_Capacity_MWe + Efficiency_Percentage + Duration +
Average_Inflation_Percentage +
    Average_Concrete_Price
  Res.Df    RSS Df Sum of Sq      F                Pr(>F)
1    260 1425.9
2    229  660.8 31    765.14 8.5535 < 0.00000000000000022 ***
```

- Since the $\mathrm{p-value}$ of the ANOVA table is <0.05, it means that the **null hypotheses has to be rejected**. This means that **at least one independent variable in our model is useful junk.**

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Statistical Analysis (Step-wise Regression Procedure to find most significant variable)

- **Pent =0.1 and Prem = 0.3.**
- As per the result of Step-wise regression procedure the **most significant variable** is **Gross_Capacity_Mwe.**
- While the significant variables are **Duration, Average_Concrete_Price, Reactor_Type and Country.**

```
                                          Pr(>|t|)
(Intercept)                               0.039787 *
Gross_Capacity_MWe                      0.00000684 ***
Duration                                0.00004660 ***
factor(Reactor_Type)FBR                   0.000355 ***
factor(Reactor_Type)HTGR                0.00003420 ***
factor(Reactor_Type)LWGR                  0.003693 **
factor(Country)BELARUS                    0.002864 **
factor(Country)IRAN, ISLAMIC REPUBLIC OF  0.014372 *
Average_Concrete_Price                  0.00001139 ***

Residual standard error: 1.691 on 231 degrees of freedom
Multiple R-squared:  0.5366,    Adjusted R-squared:  0.4784
F-statistic: 9.223 on 29 and 231 DF,  p-value: < 0.00000000000000022
```


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Statistical Analysis (Backward Regression Procedure to find least significant variable)

- **Prem = 0.3.**
- As per the result of Step-wise regression procedure the **most in-significant variable** is **Efficiency_percentage.**
- The common significant variables, **both in Step-wise and Backward procedure** are: - are **Gross_Capacity_MWe, Duration, Average_Concrete_Price, Reactor_Type and Country.** We shall now do Individual Coefficient Test (T-test) in order to finally know our significant variables and make our first order model.

```
                                                Pr(>|t|)
(Intercept)                                     0.039787 *
factor(Country)BELARUS                          0.002864 **
factor(Country)IRAN, ISLAMIC REPUBLIC OF        0.014372 *
factor(Reactor_Type)FBR                         0.000355 ***
factor(Reactor_Type)HTGR                       0.00003420 ***
factor(Reactor_Type)LWGR                        0.003693 **
Gross_Capacity_MWe                         0.00000684 ***
Duration                                   0.00004660 ***
Average_Concrete_Price                     0.00001139 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.691 on 231 degrees of freedom
Multiple R-squared:  0.5366,    Adjusted R-squared:  0.4784
F-statistic: 9.223 on 29 and 231 DF,  p-value: < 0.00000000000000022
```


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Statistical Analysis (Individual t-test)

$$H0 : \beta i = 0$$

$$Ha : \ \beta i \neq 0 \ (i = 1,2, \ldots, 7)$$

- From our Individual **T-test**, we can see that **Gross_Capacity_MWe, Duration, Average_Concrete_Price, Reactor_Type and Country** are the **only significant variables** in our regression model, as **confirmed** by **Step-Wise** and **Backward procedure.**
- **Efficiency_Percentage** and **Average_Inflation_Percentage** variables are **insignificant variables**, therefore we shall remove them.

```
factor(Country)BELARUS                          **
factor(Country)IRAN, ISLAMIC REPUBLIC OF        *
factor(Reactor_Type)FBR                         **
factor(Reactor_Type)HTGR                        ***
factor(Reactor_Type)LWGR                        *
Gross_Capacity_MWe                              ***
Duration                                        ***
Average_Concrete_Price                          ***
```


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Model after Step-wise, Backward and T-test(Additive Model):

- Country
- Duration
- Reactor_Type
- Gross_Capacity_Mwe
- Average_Concrete_Price
- **Total: 5 predictor variables from 7 previously.**
- **Adj R2 = 0.4784**

```
new.nuclear.regression.model.reduced
        <-lm(Cost_Billion_Dollars~factor(Country)
        + factor(Reactor_Type) + Gross_Capacity_MWe + Duration
        + Average_Concrete_Price, data = nuclear.df)
```


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Statistical Analysis (Partial f-test to confirm if Average_inflation %, Efficiency % are insignificant)

$$Ho: < List\ of\ insignificant\ variables > = 0$$

- The Partial F test clearly states that we **cannot reject the NULL hypothesis**, therefore it was a **correct decision** to **remove Average_inflation % and Efficiency %**.

```
Analysis of Variance Table

Model 1: Cost_Billion_Dollars ~ factor(Country) + factor(Reactor_Type) +
    Gross_Capacity_MWe + Duration + Average_Concrete_Price
Model 2: Cost_Billion_Dollars ~ factor(Country) + factor(Reactor_Type) +
    Gross_Capacity_MWe + Efficiency_Percentage + Duration +
Average_Inflation_Percentage +
    Average_Concrete_Price
  Res.Df   RSS Df Sum of Sq      F Pr(>F)
1    231 660.8
2    229 660.8  2 0.0017806 0.0003 0.9997
```


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Statistical Analysis (Interaction Model)

$$H0 : \beta i = 0$$
$$Ha : \ \beta i \neq 0 \ (i = 1,2,\ldots,5)$$

- The result of interaction model shows that this two interaction terms which are

  Country *Reactor Type

  Country*Average Concrete Price

are significant as their p-value < 0.05, we have to **reject the NULL Hypothesis** for those variables.

Also note that the **Adjusted R-Squared of our interaction model is 0.6983** as opposed to **0.4784 for our purely additive model**. Therefore our interaction model is superior and we must accept this model.

```
factor(Country)CHINA:factor(Reactor_Type)FBR          0.02886 *
factor(Country)CHINA:Average_Concrete_Price           0.00243 **
factor(Country)RUSSIA:Average_Concrete_Price          0.00780 **

Residual standard error: 1.334 on 176 degrees of freedom
Multiple R-squared:  0.7802,    Adjusted R-squared:  0.6753
F-statistic: 7.437 on 84 and 176 DF,  p-value: < 0.00000000000000022
```


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Model after Step-wise, Backward and T-test(Interaction Model):

- Country
- Duration
- Reactor_Type
- Gross_Capacity_Mwe
- Average_Concrete_Price
- Country * Reactor_Type
- Country * Average_Concrete_Price
- **Total: 5 predictor variables from 7 previously.**
- **Adj R2 = 0.6983**

$$
\begin{aligned}
&interact.new.nuclear.regression.model.reduced.cleaned \\
&\quad < -lm(Cost_Billion_Dollars \sim factor(Country) \\
&\quad + factor(Reactor_Type) + Gross_Capacity_MWe + Duration \\
&\quad + Average_Concrete_Price + factor(Country) * factor(Reactor_Type) \\
&\quad + factor(Country) * Average_Concrete_Price \ , data \ = \ nuclear.df)
\end{aligned}
$$

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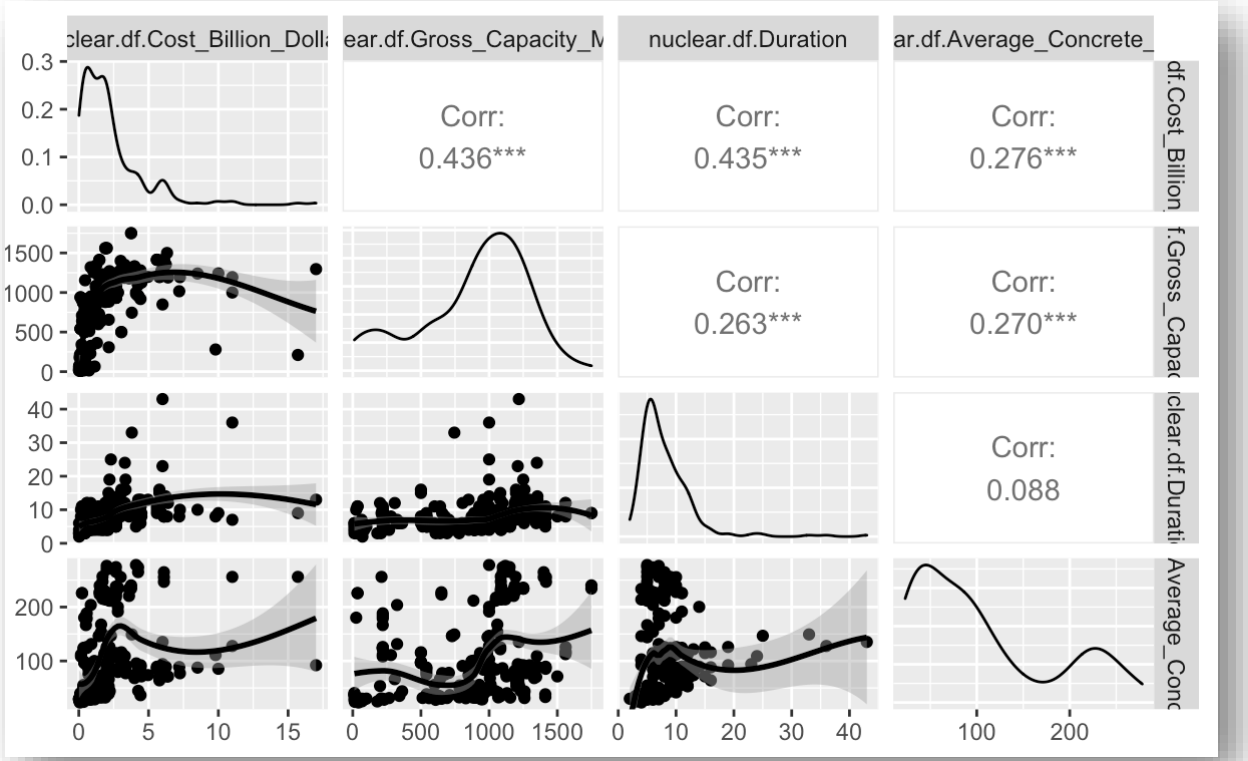
Statistical Analysis (Higher Order Model)

- In our interact model, we should start by adding Higher orders of Gross Capacity MWe because it has the highest correlation with response variable Cost . (0.436)
- (Gross_Capacity_Mwe^2) = 0.17 > 0.05 thus we fail to reject null hypothesis and the term is not significant. Similar results were observed for Duration and Average Concrete_Price.
- Therefore our **Interaction Model** is the **best model**.



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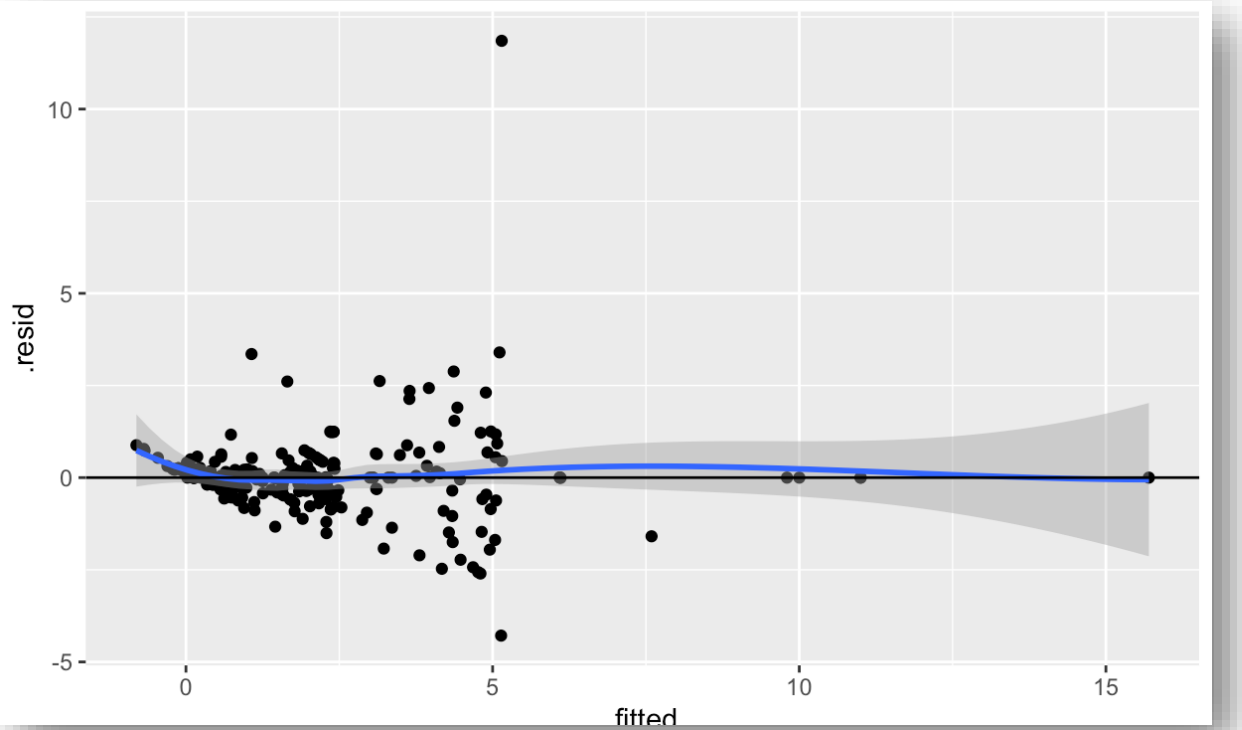
Statistical Analysis (Linearity Assumption Test)

- We plotted the **residual vs predicted** value for **Interaction model** to check linearity.
- Residual plot looks almost perfectly linear as **blue line is almost linear to black line**.



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Statistical Analysis (Equal Variance Assumption)

$$H0: \ heteroscedasticity\ is\ not\ present\ (homoscedasticity)$$
$$Ha: \ heteroscedasticity\ is\ present$$

- Using Breusch-Pagan in order to check for heteroscedasticity.

```
        studentized Breusch-Pagan test

data:  nuclear.best.model
BP = 19.912, df = 53, p-value = 1
```

- As we can see from the BP Test, the **p-value > 0.05**. Therefore, **we fail to reject null hypothesis**.
- This means that in our best fit model, **the error terms have a constant variance i.e. homoscedasticity.**

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Statistical Analysis (Normality Assumption)

$$H0: \textit{ the sample data are significantly normally distributed}$$
$$Ha: \textit{ the sample data are not significantly normally distributed}$$

- Using Shapiro - Wilk test. in order to check for heteroscedasticity.

```
        Shapiro-Wilk normality test

data:  residuals(nuclear.best.model)
W = 0.67977, p-value < 0.00000000000000022
```

- we can see that the **p-value < 0.05. Therefore H0 has to be rejected.** This means that the sample data are not significantly normally distributed.

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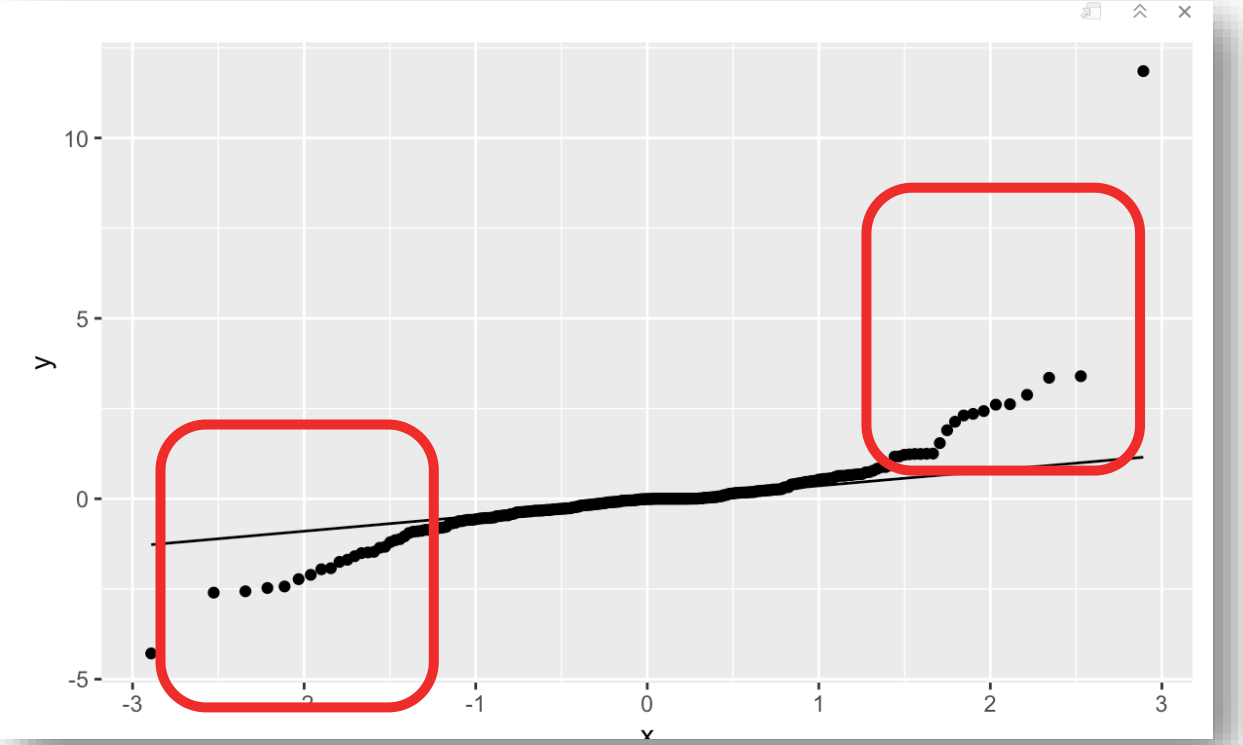
Statistical Analysis (Normality Assumption)

- Normal Q-Q Plot:

An S-shaped pattern of deviations indicates that the residuals have excessive kurtosis, i.e., there are either too many or two few large errors in both directions.



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Statistical Analysis (Normality Assumption: Box-plot transformation)

- As observed from the dataset that there is does not exist normality using box-plot transformation to normalize.
- Bestlambda = 0.3131313.

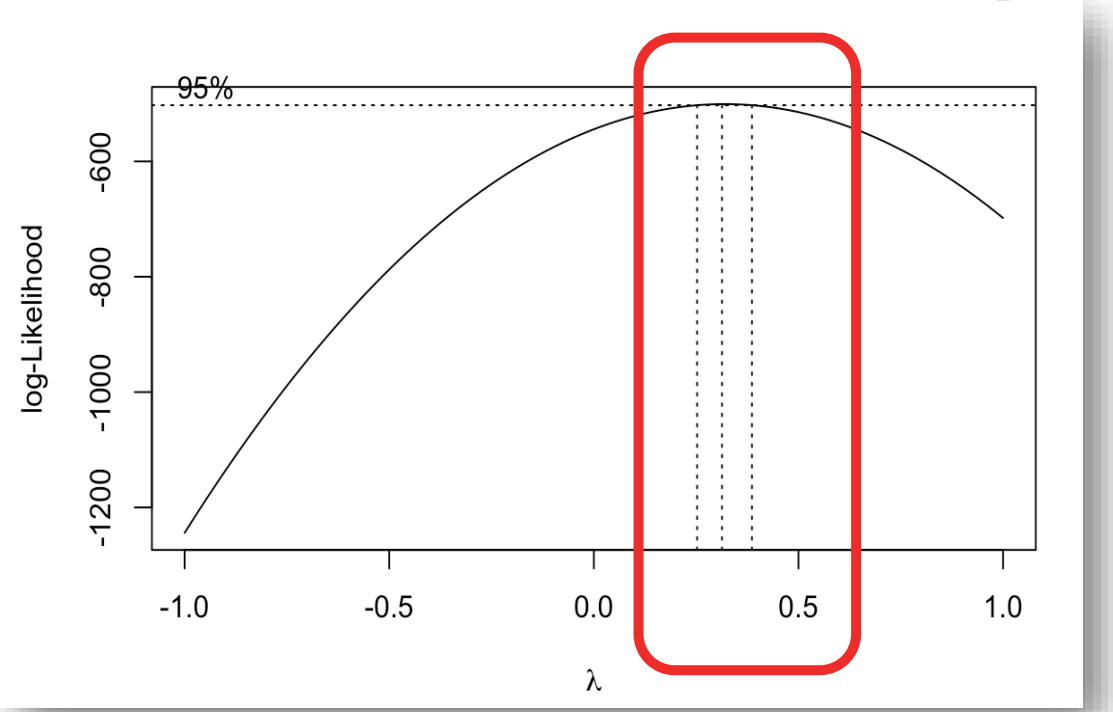
$$H0: \textit{heteroscedasticity is not present (homoscedasticity)}$$

$$Ha: \textit{heteroscedasticity is present}$$

```
        studentized Breusch-Pagan test

data:  box.cox.nuclear.best.model
BP = 48.411, df = 53, p-value = 0.6532
```

- As we can see from the BP Test, the p-value > 0.05. Therefore we fail to reject null hypothesis. This means that in our box cox fit model, the error terms have a constant variance i.e. homoscedasticity.
- Do note that the p-value decreased after the transformation.



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Statistical Analysis (Normality Assumption: After box-plot tranformation)

$$H0: \textit{the sample data are significantly normally distributed}$$

$$Ha: \textit{the sample data are not significantly normally distributed}$$

- Checking for Normality Assumption using Shapiro - Wilk test.

```
        Shapiro-Wilk normality test

data:  residuals(box.cox.nuclear.best.model)
W = 0.91051, p-value = 0.00000000002259
```

- From the Shapiro-Wilk normality test, we can see that the p-value < 0.05. Therefore, H0 has to be rejected. This means that the **sample data is still not significantly normally distributed**. Do note that the **p-value** of Shapiro-Wilk normality test has improved sharply after the transformation from **2.2e-16 to 2.2e-11**, but it is still not enough.

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Model after Step-wise, Backward and T-test(After Box-Cox Transformation Model):

- Country
- Duration
- Reactor_Type
- Gross_Capacity_Mwe
- Average_Concrete_Price
- Country * Reactor_Type
- Country * Average_Concrete_Price
- **Total: 5 predictor variables and lambda = 0.3131313**
- **Adj R2 = 0.8019**

$$box.cox.nuclear.best.model < -lm((((Cost_Billion_Dollars\text{^}0.3131313) - 1)/0.3131313) \sim factor(Country) + factor(Reactor_Type) + Gross_Capacity_MWe + Duration + Average_Concrete_Price + factor(Country) * factor(Reactor_Type) + factor(Country) * Average_Concrete_Price, data = nuclear.df)$$

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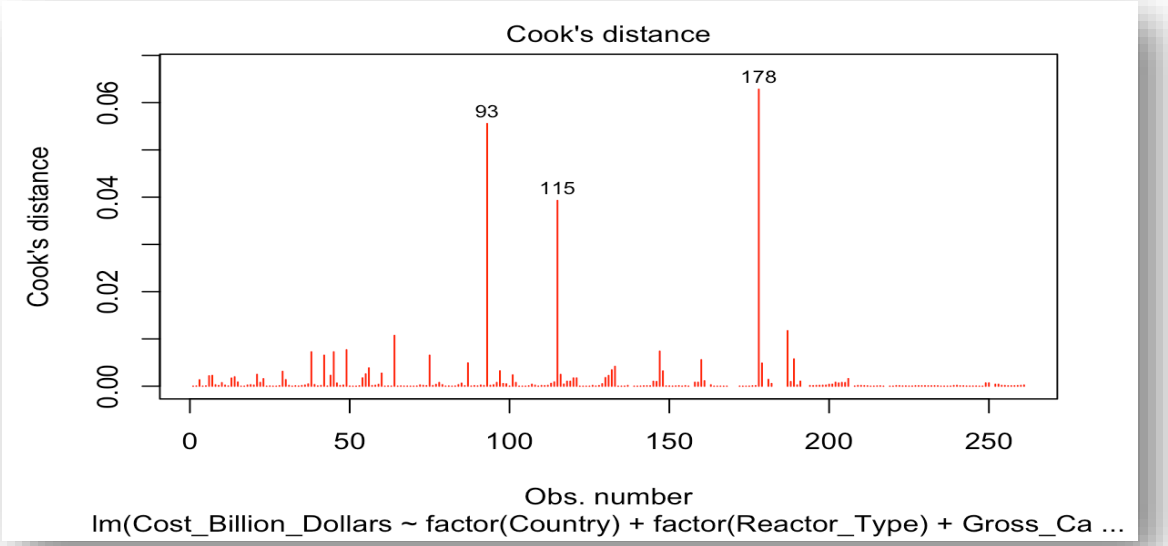
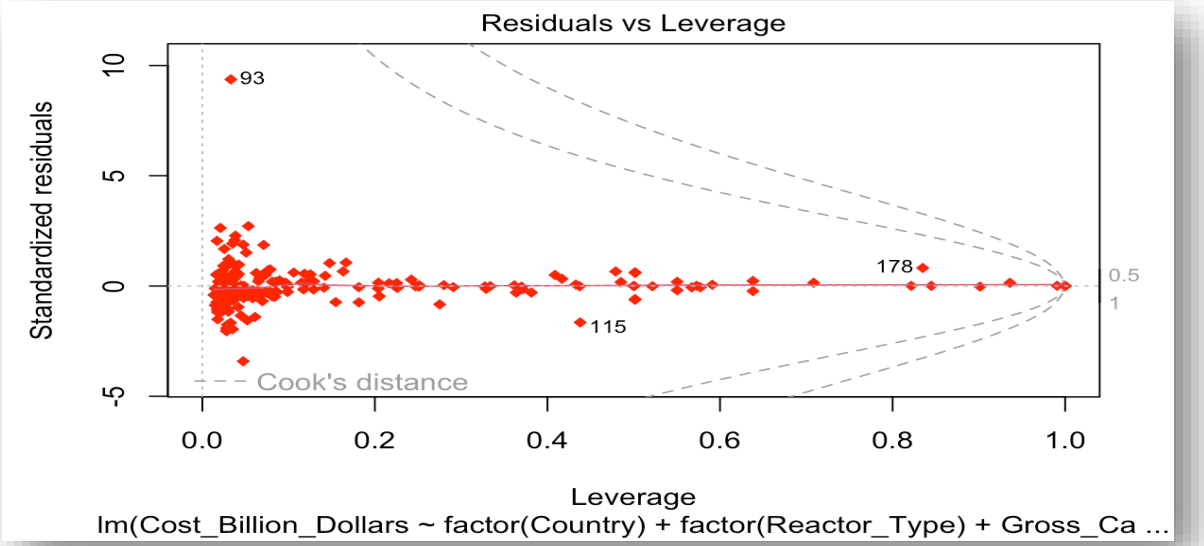
Statistical Analysis (Outliers)





- There is not a single data point whose cook's distance is > 0.5. Hence according to Cook, no outlier detected.
- Similarly for the Residuals vs Leverage plot **no outliers found**

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Overall Conclusion of our Project

- Following terms stand significant to the COST of a nuclear plant :

  Country , Duration, Reactor_Type, Gross_Capacity_Mwe, Average_Concrete_Price,
  Country * Reactor_Type, Country * Average_Concrete_Price

- **Adj R2 = 0.8019 our final model.**
- Initially we considered each factor will play a significant role to the cost variable, but it turns out only half of them are statically significant.

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